

DATA STORYTELLING & STATISTICAL VALIDATION

Sales Data Analysis – Business Insights

Project Focus	Tools Used
E-Commerce Sales Performance	Power BI + Python

1. Problem Statement

The company has a large amount of sales data, but raw data alone does not clearly explain business performance. The company needs to identify sales trends, top-performing products and categories, high-performing cities, and customer purchasing patterns to support better business decisions.

2. Business Objective

To analyze sales data and transform it into meaningful insights that can help the company understand sales performance, identify growth opportunities, and make data-driven decisions.

3. Analysis Performed

- Total Sales
- Product Performance
- Category Performance
- City-wise Sales
- Monthly Sales Trends
- Customer Purchasing Patterns
- Gender-based Sales

Tools Used: Power BI for visualization and dashboard analysis; Python for statistical validation and hypothesis testing.

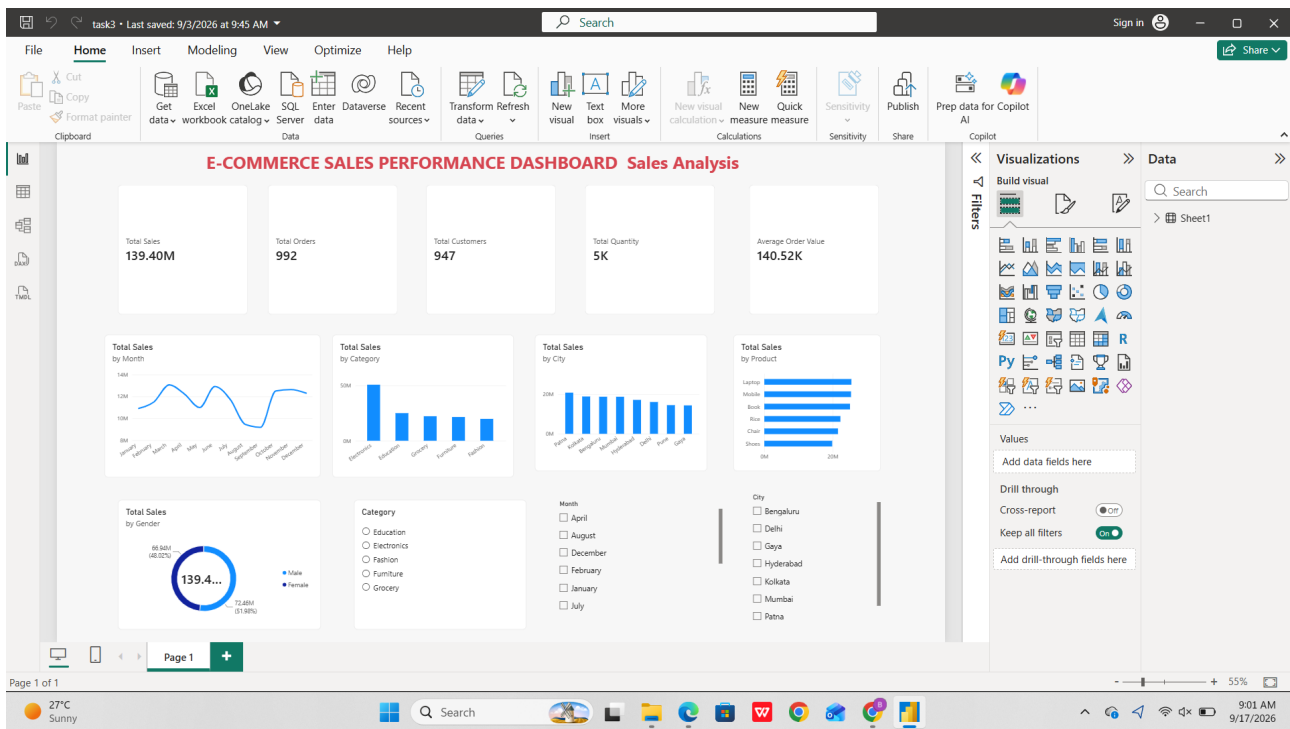
4. Key Insights from the Dataset

Metric	Result	Business Insight
Highest Category	Electronics ₹50,778,581.70	Electronics contributes the largest share of total revenue.
Highest Product	Laptop ₹25,443,008.51	Laptops generate the highest product revenue.
Highest Month	March ₹13,059,899.94	March recorded the highest monthly sales.
Highest City	Patna ₹20,826,584.43	Patna recorded the highest sales among the analyzed cities.

These insights are based on the sales results shown in the Power BI analysis supplied for Task 4.

5. Power BI Sales Dashboard

The dashboard summarizes the main sales KPIs and visualizes performance by month, category, city, product, and gender.



6. Hypothesis Testing

Business Question

Is there a significant difference in average Total Sales between Male and Female customers?

Null Hypothesis (H_0)

There is no significant difference in average Total Sales between Male and Female customers.

Alternative Hypothesis (H_1)

There is a significant difference in average Total Sales between Male and Female customers.

Statistical Test

Independent Samples T-Test (Welch's t-test) was used to compare the average Total Sales of the two gender groups.

Significance Level

$\alpha = 0.05$

Group	Number of Records	Average Total Sales
Female	489	₹136,883.21
Male	511	₹141,807.34

7. Statistical Results & Validation

Test Measure	Result
T-Statistic	0.682610
P-Value	0.495011
Significance Level	0.05
Decision	Fail to Reject H_0

Decision Rule

If $p\text{-value} < 0.05$, reject H_0 . If $p\text{-value} \geq 0.05$, fail to reject H_0 .

Interpretation

The $p\text{-value}$ is 0.495011, which is greater than 0.05. Therefore, the null hypothesis is not rejected.

Statistical Conclusion

There is no statistically significant difference in average Total Sales between Male and Female customers in this dataset. Although the observed average for Male customers is higher, the difference is not statistically significant at the 5% significance level.

8. Overall Business Story

The analysis shows strong sales performance during 2025. Electronics was the highest-performing category, Laptop was the top-performing product, Patna recorded the highest city-wise sales, and March recorded the highest monthly sales. These findings help the company identify its strongest products, markets, and sales periods.

Overall Business Performance

Total Records	Unique Orders	Unique Customers
1,000	992	947
Total Sales	Analysis Period	Top Category
₹139,399,439.65	2025	Electronics

9. Recommendations / Call to Action

- Focus on the Electronics category because it generated the highest revenue.
- Maintain sufficient Laptop inventory to meet observed demand.
- Study the reasons for high March sales and apply successful strategies to other periods where appropriate.
- Strengthen marketing and customer engagement in Patna based on its high sales contribution.
- Review low-performing products and categories to identify opportunities for improvement.
- Use multiple customer and sales factors—not gender alone—when developing business strategies.

Call to Action: Use the dashboard insights and statistical evidence to support inventory planning, marketing decisions, and ongoing sales-performance monitoring.